

Biomarker Distributions

METIMMOX-1 Economic Evaluation

Ben Geisler

2026-02-22

Table of contents

1	Introduction	1
2	Sample Characteristics	2
3	CRP Stratification	2
4	TLR Stratification	3
5	TMB/BRAF Stratification	3
6	Biomarker Correlations	4

1 Introduction

This report presents the distribution of patient characteristics stratified by treatment arm and biomarker status. Three biomarker strategies are evaluated:

- **CRP**: C-reactive protein (positive: $\text{CRP} < 5$)
- **TLR**: Tumor lesion reduction (positive: $\text{TLR} \geq 10\%$)
- **TMB/BRAF**: Combined biomarker (positive: $\text{TMB} \geq 9 \text{ mut/MB}$ OR BRAF mutation)

Each table shows the distribution of age, sex, and the other two biomarkers (both categorical and continuous) within subgroups defined by treatment arm and biomarker status.

2 Sample Characteristics

Table 1: Sample Characteristics

Characteristic	N (%)
Total patients (complete cases)	65
Control arm (FLOX alone)	29 (44.6%)
Experimental arm (alternating FLOX/nivolumab)	36 (55.4%)
Deaths (OS events)	56 (86.2%)
Progressions (PFS events)	48 (73.8%)
CRP-positive (low CRP)	22 (33.8%)
TLR-positive (early response)	41 (63.1%)
TMB/BRAF-positive	29 (44.6%)

3 CRP Stratification

Distribution of patient characteristics by treatment arm and CRP status.

Table 2: Patient Characteristics by Treatment Arm and CRP Status

	level	Control / -	Control / +	Exp / -	Exp / +
n		24	19	5	17
Age (mean (SD))		66.29 (10.20)	60.74 (8.99)	58.60 (12.78)	65.94 (9.24)
sex (%)	Male	15 (62.5)	10 (52.6)	2 (40.0)	8 (47.1)
	Female	9 (37.5)	9 (47.4)	3 (60.0)	9 (52.9)
TLR_continuous (mean (SD))		0.78 (0.22)	1.04 (0.38)	0.65 (0.12)	0.81 (0.21)
tlr_status (%)	TLR-	7 (29.2)	13 (68.4)	0 (0.0)	4 (23.5)
	TLR+	17 (70.8)	6 (31.6)	5 (100.0)	13 (76.5)
TMB_continuous (mean (SD))		7.03 (3.12)	5.87 (3.98)	6.60 (2.97)	8.60 (2.46)
tmb_braf_status (%)	TMB/BRAF-	12 (50.0)	13 (68.4)	4 (80.0)	7 (41.2)
	TMB/BRAF+	12 (50.0)	6 (31.6)	1 (20.0)	10 (58.8)
Death_event (%)	No	3 (12.5)	0 (0.0)	2 (40.0)	4 (23.5)
	Yes	21 (87.5)	19 (100.0)	3 (60.0)	13 (76.5)
Progression_event (%)	No	7 (29.2)	2 (10.5)	3 (60.0)	5 (29.4)
	Yes	17 (70.8)	17 (89.5)	2 (40.0)	12 (70.6)

4 TLR Stratification

Distribution of patient characteristics by treatment arm and TLR status.

Table 3: Patient Characteristics by Treatment Arm and TLR Status

	level	Control / -	Control / +	Exp / -	Exp / +
n		7	17	22	19
Age (mean (SD))		68.71 (8.62)	59.59 (9.83)	63.77 (11.37)	66.42 (7.82)
sex (%)	Male	5 (71.4)	8 (47.1)	12 (54.5)	10 (52.6)
	Female	2 (28.6)	9 (52.9)	10 (45.5)	9 (47.4)
CRP_continuous (mean (SD))		18.43 (13.45)	15.44 (16.22)	12.05 (13.30)	4.71 (7.45)
crp_status (%)	CRP-	7 (100.0)	13 (76.5)	17 (77.3)	6 (31.6)
	CRP+	0 (0.0)	4 (23.5)	5 (22.7)	13 (68.4)
TMB_continuous (mean (SD))		7.33 (4.72)	6.58 (3.91)	6.85 (2.56)	7.68 (3.28)
tmb_braf_status (%)	TMB/BRAF-	3 (42.9)	10 (58.8)	13 (59.1)	10 (52.6)
	TMB/BRAF+	4 (57.1)	7 (41.2)	9 (40.9)	9 (47.4)
Death_event (%)	No	0 (0.0)	0 (0.0)	5 (22.7)	4 (21.1)
	Yes	7 (100.0)	17 (100.0)	17 (77.3)	15 (78.9)
Progression_event (%)	No	1 (14.3)	2 (11.8)	9 (40.9)	5 (26.3)
	Yes	6 (85.7)	15 (88.2)	13 (59.1)	14 (73.7)

5 TMB/BRAF Stratification

Distribution of patient characteristics by treatment arm and TMB/BRAF status.

Table 4: Patient Characteristics by Treatment Arm and TMB/BRAF Status

	level	Control / -	Control / +	Exp / -	Exp / +
n		16	20	13	16
Age (mean (SD))		64.19 (9.51)	63.00 (10.30)	65.92 (12.62)	63.44 (8.36)
sex (%)	Male	9 (56.2)	11 (55.0)	8 (61.5)	7 (43.8)
	Female	7 (43.8)	9 (45.0)	5 (38.5)	9 (56.2)
CRP_continuous (mean (SD))		13.03 (14.80)	10.66 (13.68)	14.28 (11.96)	8.66 (13.29)
crp_status (%)	CRP-	12 (75.0)	13 (65.0)	12 (92.3)	6 (37.5)
	CRP+	4 (25.0)	7 (35.0)	1 (7.7)	10 (62.5)
TLR_continuous (mean (SD))		0.73 (0.19)	0.98 (0.36)	0.80 (0.24)	0.86 (0.27)
tlr_status (%)	TLR-	3 (18.8)	10 (50.0)	4 (30.8)	7 (43.8)
	TLR+	13 (81.2)	10 (50.0)	9 (69.2)	9 (56.2)
Death_event (%)	No	2 (12.5)	0 (0.0)	3 (23.1)	4 (25.0)
	Yes	14 (87.5)	20 (100.0)	10 (76.9)	12 (75.0)
Progression_event (%)	No	5 (31.2)	2 (10.0)	5 (38.5)	5 (31.2)
	Yes	11 (68.8)	18 (90.0)	8 (61.5)	11 (68.8)

6 Biomarker Correlations

Table 5: Biomarker Correlations (Cramer's V)

Biomarker Pair	Cramer's V	Interpretation
CRP vs TLR	0.278	Weak
CRP vs TMB/BRAF	0.077	Negligible
TLR vs TMB/BRAF	0.019	Negligible

Note: Cramer's V interpretation: <0.1 = Negligible, $0.1-0.3$ = Weak, >0.3 = Moderate/Strong. The observed correlations are weak to negligible.

Report completed on: 2026-02-22

Repository: ben-geisler/METIMMOX-1

Report version: 1.1